

ESTIMATION OF AVERAGE GRAIN FROM MICROSTRUCTURE IMAGES

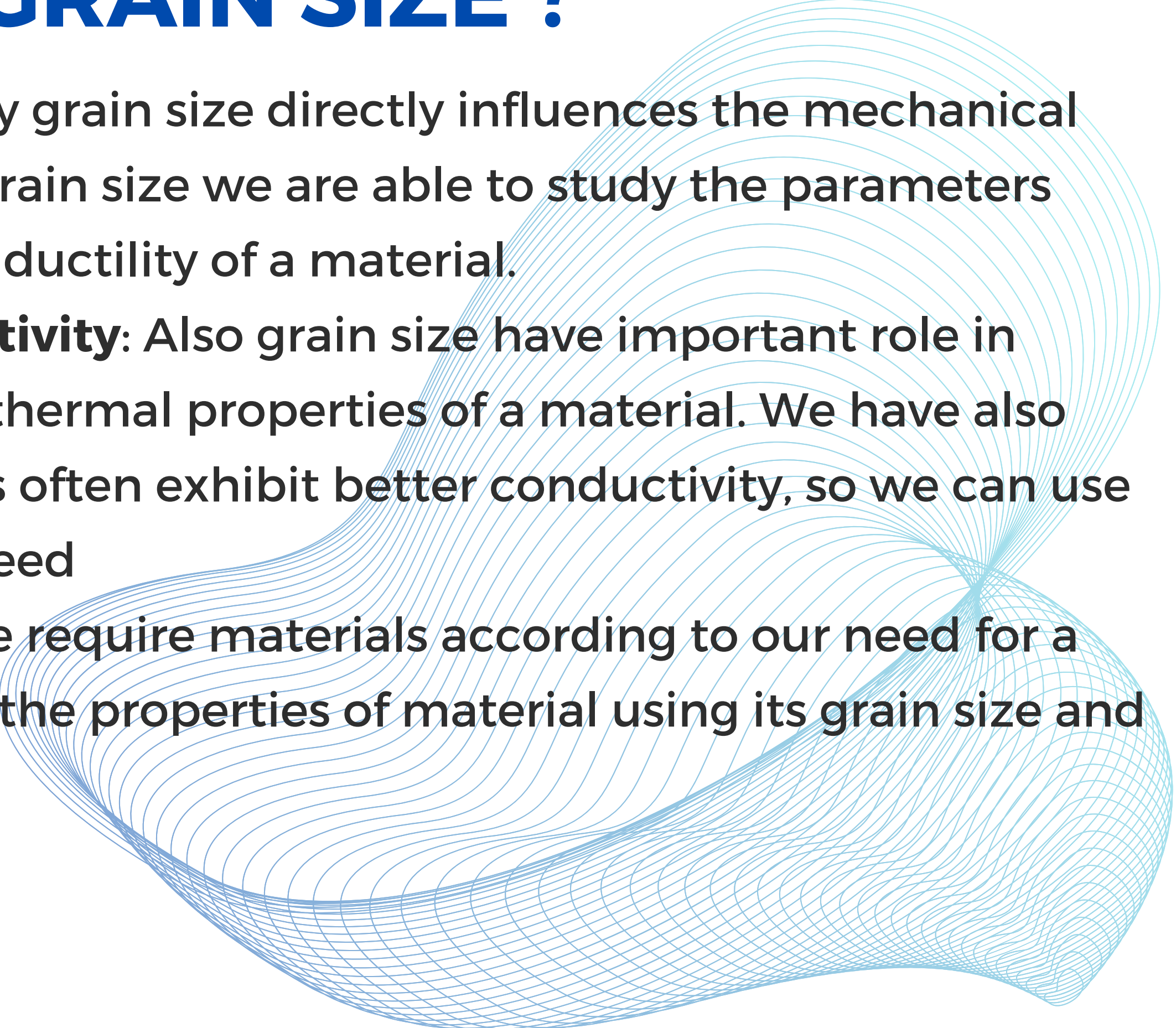
USING PYTHON AND
MACHINE LEARNING

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WHY AVERAGE GRAIN SIZE ?

- **Mechanical Properties:** Basically grain size directly influences the mechanical properties of materials. Using grain size we are able to study the parameters such as strength, hardness, and ductility of a material.
 - **Electrical and Thermal Conductivity:** Also grain size have important role in determining the electrical and thermal properties of a material. We have also seen that fine-grained materials often exhibit better conductivity, so we can use the material according to our need
 - **Material Design:** many times we require materials according to our need for a specific objective so inspecting the properties of material using its grain size and properties will be more helpful
- 

TRADITIONAL METHODS

- **LINEAR INTERCEPT METHOD :** ON MICROGRAPHS, RANDOM LINES ARE DRAWN, AND THE NUMBER OF TIMES THESE LINES CROSS GRAINS IS RECORDED. THE AVERAGE GRAIN SIZE CAN BE DETERMINED BY MEASURING THE INTERCEPTED LINES' LENGTHS AND USING A MATHEMATICAL CALCULATION.
- **AREA FRACTION METHOD :** BY OVERLAYING A GRID ON MICROGRAPHS, THIS TECHNIQUE COUNTS THE NUMBER OF GRAINS THAT CROSS THE GRIDLINES. THE AVERAGE GRAIN SIZE CAN BE CALCULATED USING THE NUMBER OF GRAINS AND THE SIZE OF THE GRID.

NEW APPROACH

- **Machine Learning:**

Using machine learning techniques to do automatic analysis of grain size

- **Image processing :**

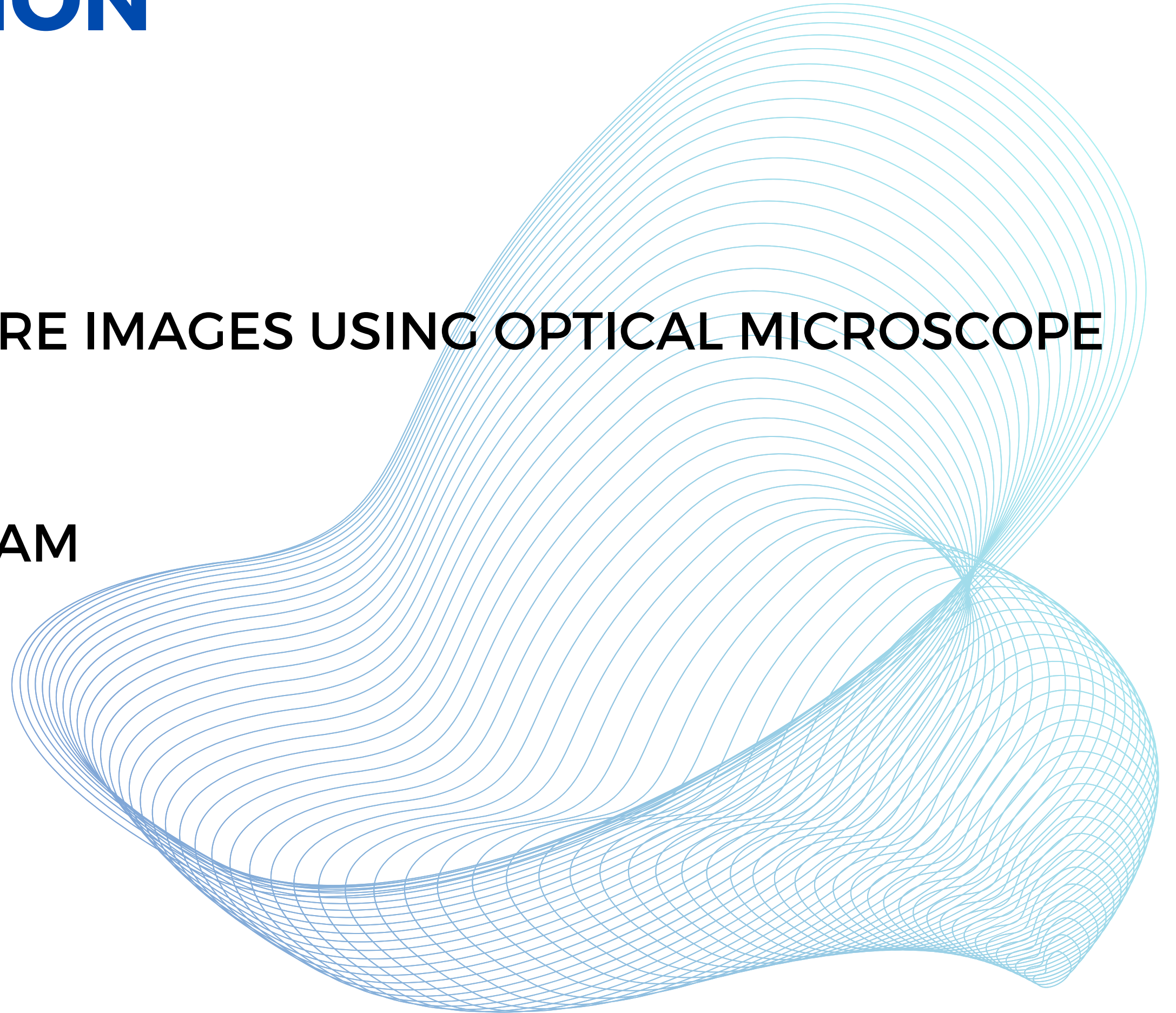
The use of digital image processing methods and software systems allows for automatic analysis of micrographs to determine grain sizes, offering a quicker and more precise alternative to manual processes.

- **X-ray Diffraction (XRD) with Synchrotron Radiation of High Energy:**

Using advanced XRD methods with high-energy synchrotron radiation to get better clarity when figuring out the size of grains

MODEL DESCRIPTION

1. PREPARATION OF SAMPLE
2. CAPTURING MICROSTRUCTURE IMAGES USING OPTICAL MICROSCOPE
3. CAPTURING FESEM IMAGES
4. MAKING OF OPENCV PROGRAM

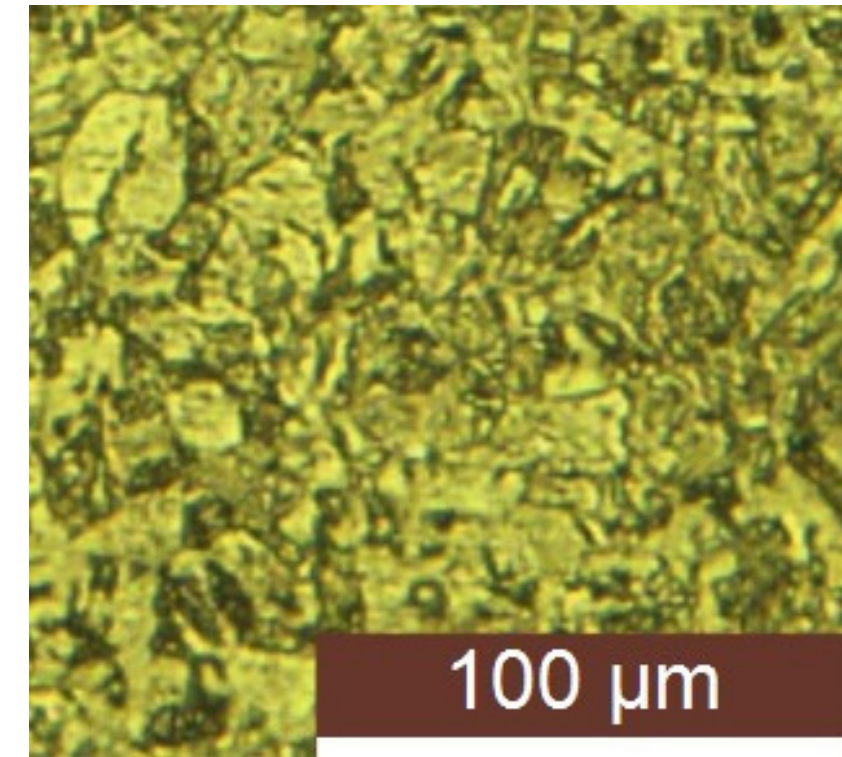
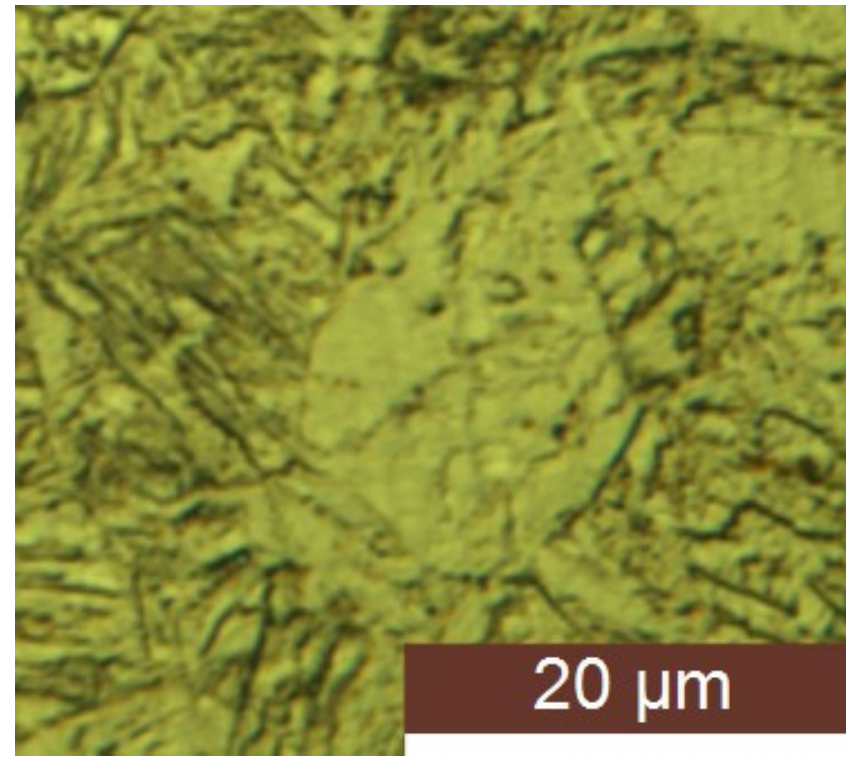
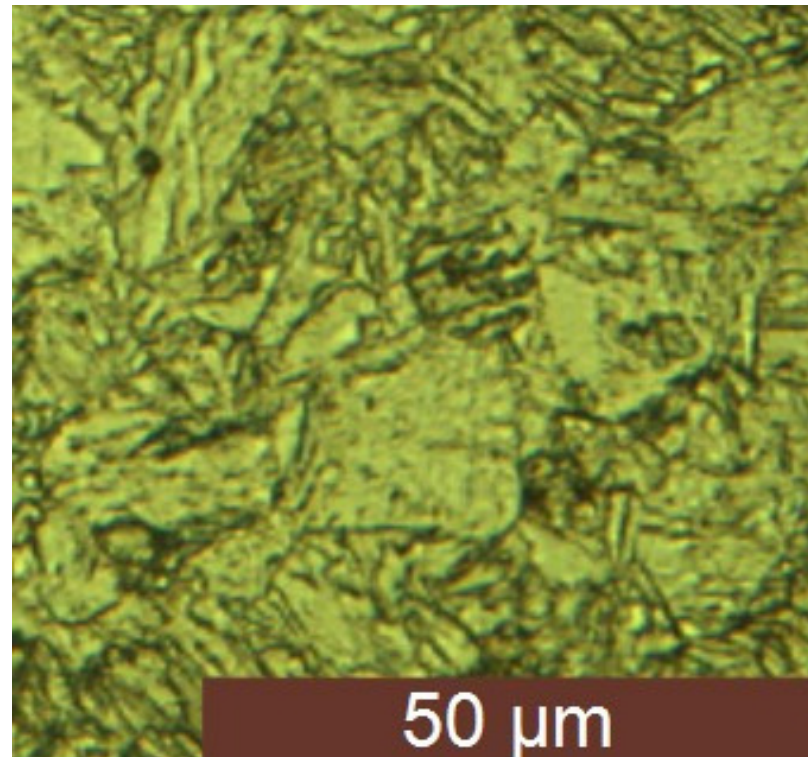


SAMPLE PREPARATION

- 1. P91 STEEL, A ROBUST CHROME-MOLY ALLOY, UNDERGOES SPECIFIC HEAT TREATMENT FOR ENHANCED STRENGTH AND TEMPERATURE RESISTANCE.**
- 2. THE PROCESS INVOLVES NORMALIZATION AT 1050 °C, AIR COOLING TO 200 °C, AND TEMPERING AT 760 °C, CRUCIAL FOR P91'S CREEP STRENGTH.**
- 3. P91'S COMPOSITION INCLUDES 9% CHROMIUM, 1% MOLYBDENUM, AND VANADIUM, BOOSTING OXIDATION RESISTANCE AND HIGH-TEMPERATURE CREEP STRENGTH.**
- 4. TEN P91 SAMPLES WERE CUT, NORMALIZED AT VARYING TEMPERATURES, AND THEN TEMPERED AT 760 °C FOR MICROSTRUCTURE ANALYSIS.**
- 5. SAMPLES WERE POLISHED, ETCHED WITH VILELLA AND NITAL, WITH OPTIMAL CONDITIONS IDENTIFIED AS VILELLA FOR 60S FOR GRAIN BOUNDARY ANALYSIS.**
- 6. THE PREPARED SAMPLES ARE NOW SET FOR MICROSTRUCTURE IMAGE CAPTURE AND SUBSEQUENT ANALYSIS USING A PYTHON PROGRAM.**

CAPTURING IMAGES USING MICRO-STRUCTURES IMAGES

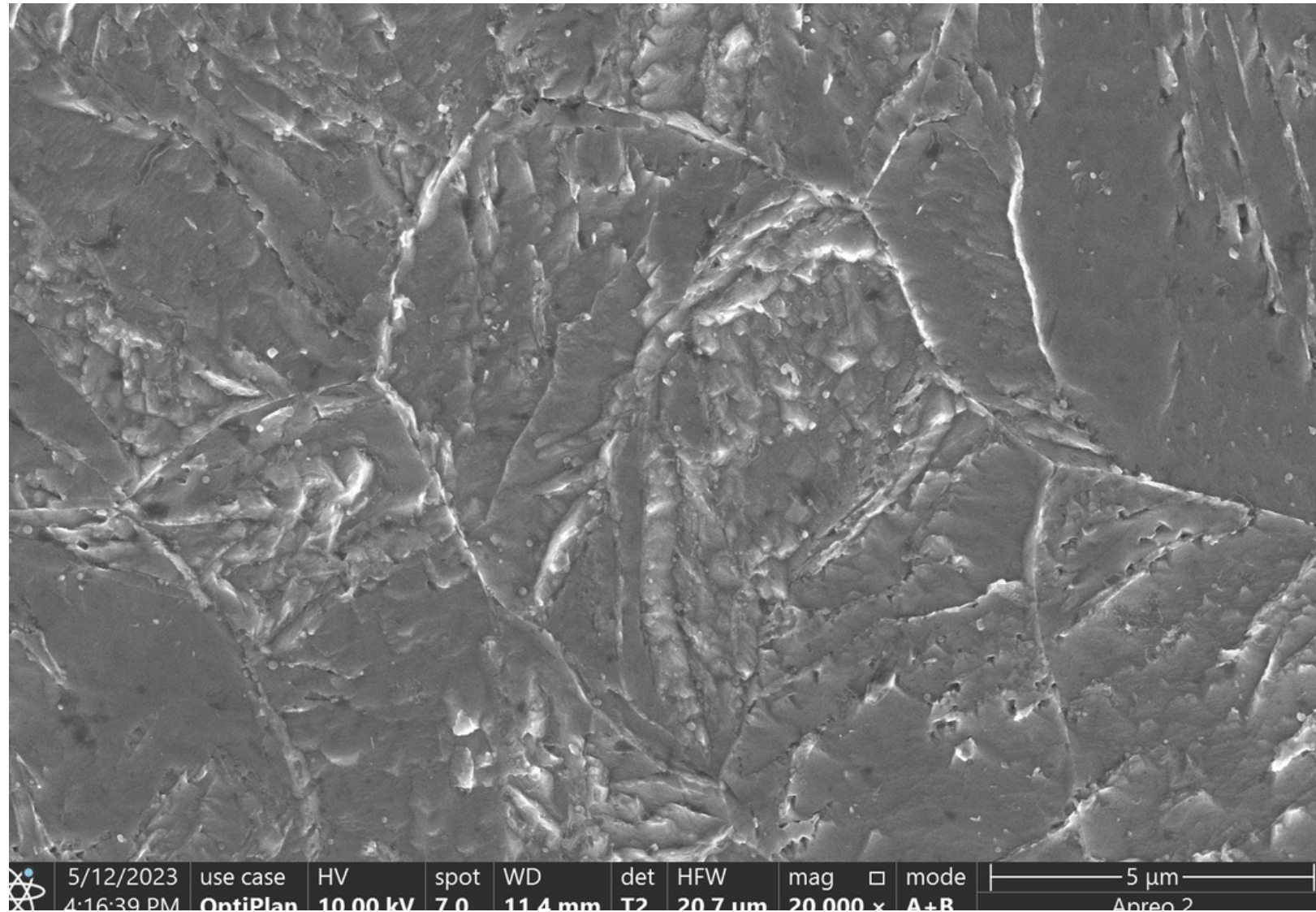
- **THE POLISHED AND READY SAMPLES (10 IN NUMBER) WERE TAKEN FIRST TO OPTICAL MICROSCOPE .**
- **THE MICROSCOPE USED WAS LECIA DM1750 M**
- **MICROSTRUCTURES IMAGES WERE TAKEN FOR ALL THE SAMPLES UNDER VARIOUS MAGNIFICATIONS OF 20X , 50X AND 100X AT VARIOUS LOCATIONS ON THE SAMPLES.**



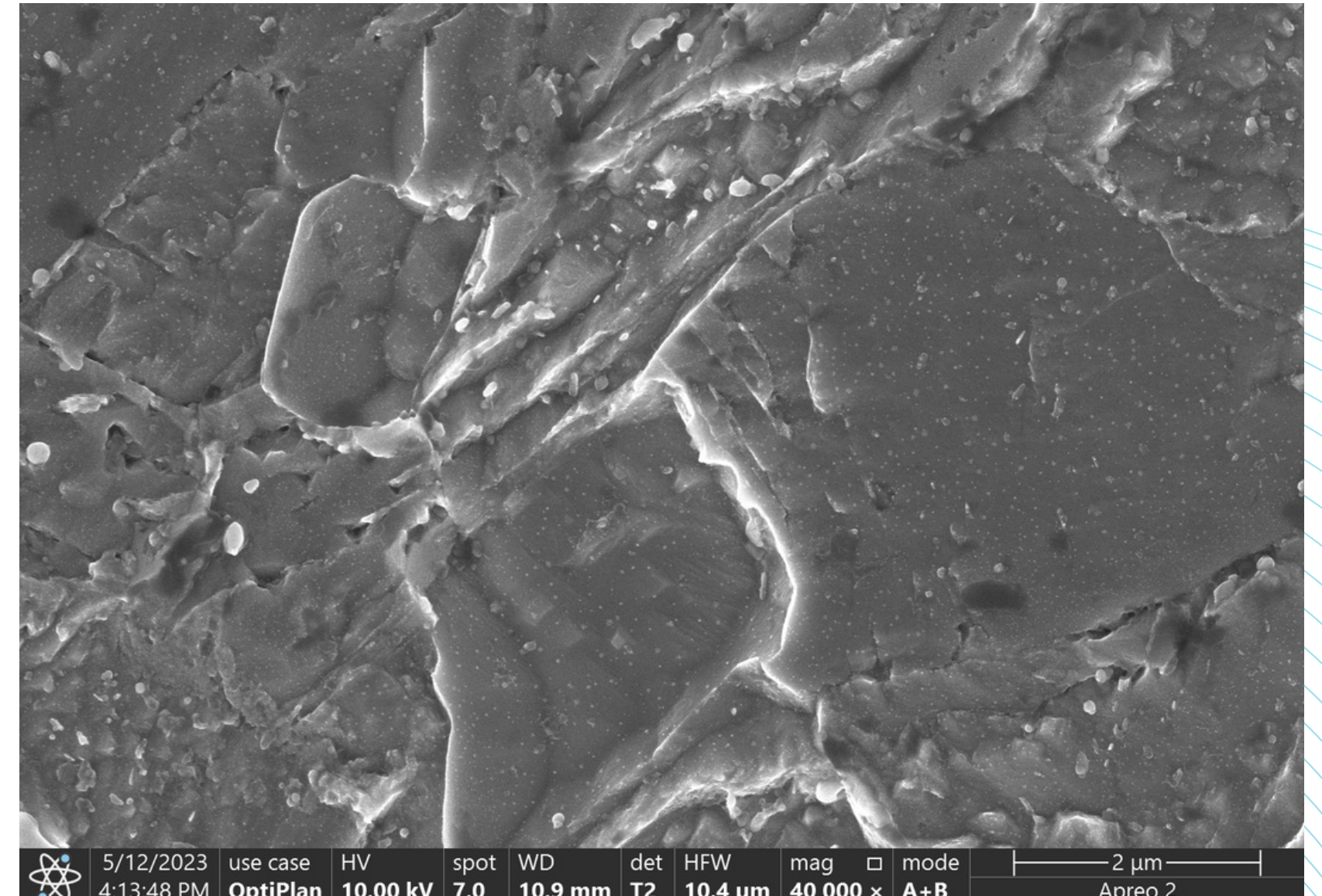
VARIOUS MICROSTRUCTURE IMAGES USED IN THIS STUDY.

CAPTURING IMAGES USING FESEM

- OPTICAL IMAGES SERVED AS AN INITIAL STEP
- SAMPLES WERE MADE CONDUCTIVE THROUGH GOLD COATING.
- SYSTEMATIC NUMBERING FACILITATED ORGANIZATION
- LOADING INTO FESEM PRECEDED PARAMETER OPTIMIZATION
- PRECISE ELECTRON BEAM POSITIONING AND SITE FOCUSING WERE CRUCIAL FOR TARGETED IMAGING.
- IMAGES WERE CAPTURED AT OPTIMIZED SETTINGS, ENSURING HIGH QUALITY.



FESEM image of N900



FESEM image of T900

VARIOUS FESEM IMAGES USED IN THIS STUDY.

SNIPPETS OF CODE

```
# In[ ]:

import numpy as np
import cv2
from matplotlib import pyplot as plt
from skimage import io,color,measure
from scipy import ndimage
img=cv2.imread(r"E:\\Second year\\SEM 2\\DC\\New-20230513T062323Z-0
cv2.imshow('image',img)

pixels_to_um=0.5
cropped_img=img[0:300,:]

#step 2
plt.hist(img.flat, bins=200,range=(0,255))
```

```
# In[ ]:

ret,thresh=cv2.threshold(img,0,255,cv2.THRESH_BINARY+cv2.THRESH_OTSU)
```

```
# In[ ]:

kernel=np.ones((3,3),np.uint8)
eroded=cv2.erode(thresh,kernel,iterations=1)
dilated=cv2.dilate(eroded,kernel,iterations =1)
#cv2.imshow("Threshold Image",thresh)
#cv2.imshow("Dilated Image",dilated)
cv2.waitKey(0)

#converting to binary
mask=dilated==255
#io.imshow(mask[250:280,250:280])

#step4
s=[[1,1,1],[1,1,1],[1,1,1]]
labeled_mask,num_labels=ndimage.label(mask,structure=s)
img2=color.label2rgb(labeled_mask,bg_label=0)
cv2.imshow("coloured labels",img2)
#cv2.waitKey(0)
```

SNIPPETS OF CODE

```
# In[ ]:

clusters=measure.regionprops(labeled_mask,img)

# In[ ]:

print(clusters[0].perimeter)

# In[ ]:

#step6
propList = ['Area',
            'equivalent_diameter', #Added... verify if it works
            'orientation', #Added, verify if it works. Angle btwn x-axis and major axis.
            'MajorAxisLength',
            'MinorAxisLength',
            'Perimeter',
            'MinIntensity',
            'MeanIntensity',
            'MaxIntensity']
```

Measuring the properties of each grain

```
# In[ ]:

output_file=open('image_measurements.csv','w')
output_file.write(','+'", "'.join(propList)+'\n')

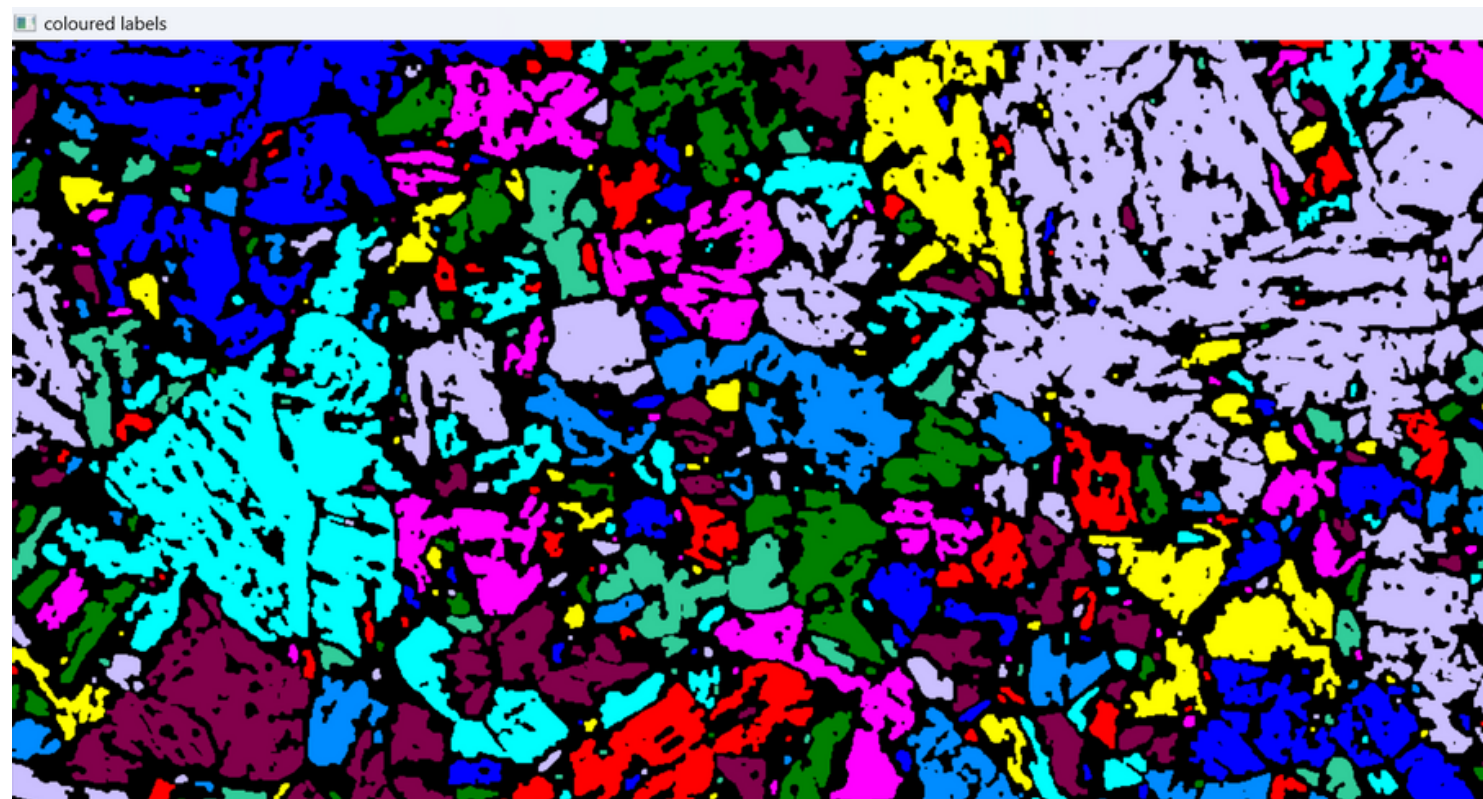
# In[ ]:

for cluster_props in clusters:
    #output cluster properties to the excel file
    output_file.write(str(cluster_props['Label']))
    for i,prop in enumerate(propList):
        if(prop == 'Area'):
            to_print = cluster_props[prop]*pixels_to_um**2 #Convert pixel square to um square
        elif(prop == 'orientation'):
            to_print = cluster_props[prop]*57.2958 #Convert to degrees from radians
        elif(prop.find('Intensity') < 0): # Any prop without Intensity in its name
            to_print = cluster_props[prop]*pixels_to_um
        else:
            to_print = cluster_props[prop] #Reamining props, basically the ones with Intensity
    output_file.write(',' + str(to_print))
    output_file.write('\n')
```

Generating the csv file from the program

WORKING OF THE PROGRAM

1. IN THIS PROGRAMME, THE IMAGES PRODUCED BY THE OPTICAL MICROSCOPE ARE UTILISED.
IMAGE PROCESSING IS THE FOUNDATION OF THIS PROGRAMME.
2. THIS CODE MAKES USE OF LIBRARIES LIKE MATPLOTLIB, SKIIMAGE, AND SCIPY.
3. SCALE OF 0.5 IS APPLIED TO CONVERT PICTURE PIXELS TO MICROMETRES.
4. THE LOCATION OF THE PIXEL WITH THE MOST GRAINS IS INDICATED BY A HISTOGRAM.
5. THE BINARY THRESHOLD METHOD IS USED TO PERFORM THE IMAGE THRESHOLDING.
6. ADDITIONALLY, EROSION AND DILATION ARE CARRIED OUT SO THAT THE GRAIN STRUCTURE'S BOUNDARIES CAN BE FILLED UTILISING THE PROGRAMME.
7. THE REGIONPROPS FUNCTION CALCULATES STATISTICS FOR EACH ITEM (CONNECTED COMPONENT) IN AN IMAGE, INCLUDING AREA, CENTROID, AND BOUNDING BOX.
8. FINALLY, THE PROGRAM GENERATES THE NECESSARY CSV FILE AND SAVES IT TO THE SYSTEM.
9. THE DATA FOR MACHINE LEARNING IS EMPLOYED IN THE GENERATED CSV FILE, AND OUTPUT IS PRODUCED.



**FIGURE 3.A COLOURED LABEL
IMAGE**



FIGURE 3.B ERODED IMAGE

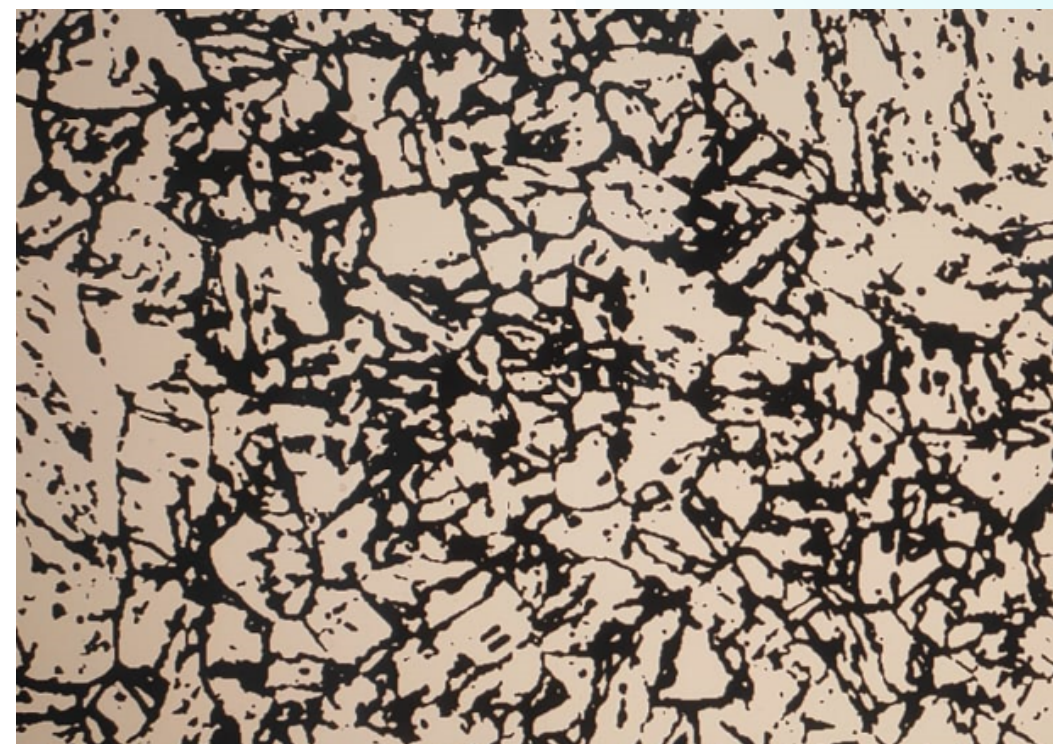


FIGURE 3.C THRESHOLD IMAGE

RESULTS AND OBSERVATIONS

- FOLLOWING ARE THE RESULTS THAT WE OBTAINED FROM THIS PROJECT BY CONSIDERING THE GRAIN SIZE:
- IMAGEJ AND A PYTHON PROGRAMME WERE USED TO LOOK AT MICROSTRUCTURE PICTURES FROM N AND T SAMPLES OF P91 STEEL.
- FOR OPTICAL PICTURES, OPENCV WAS USED, AND FOR FESEM IMAGES, IMAGEJ WAS USED.
- THE GOAL WAS TO SEE HOW WELL IMAGEJ AND THE PYTHON PROGRAMME WORKED AND HOW CONSISTENT THE RESULTS WERE.
- SPECIFICALLY LOOKED AT SAMPLE N900 (NORMALISED AT 900 °C) TO MAKE SURE IT WAS CORRECT.
- THE GOAL OF THE COMPARISON WAS TO FIND OUT IF BOTH SOFTWARE TOOLS PRODUCED SIMILAR RESULTS.

```

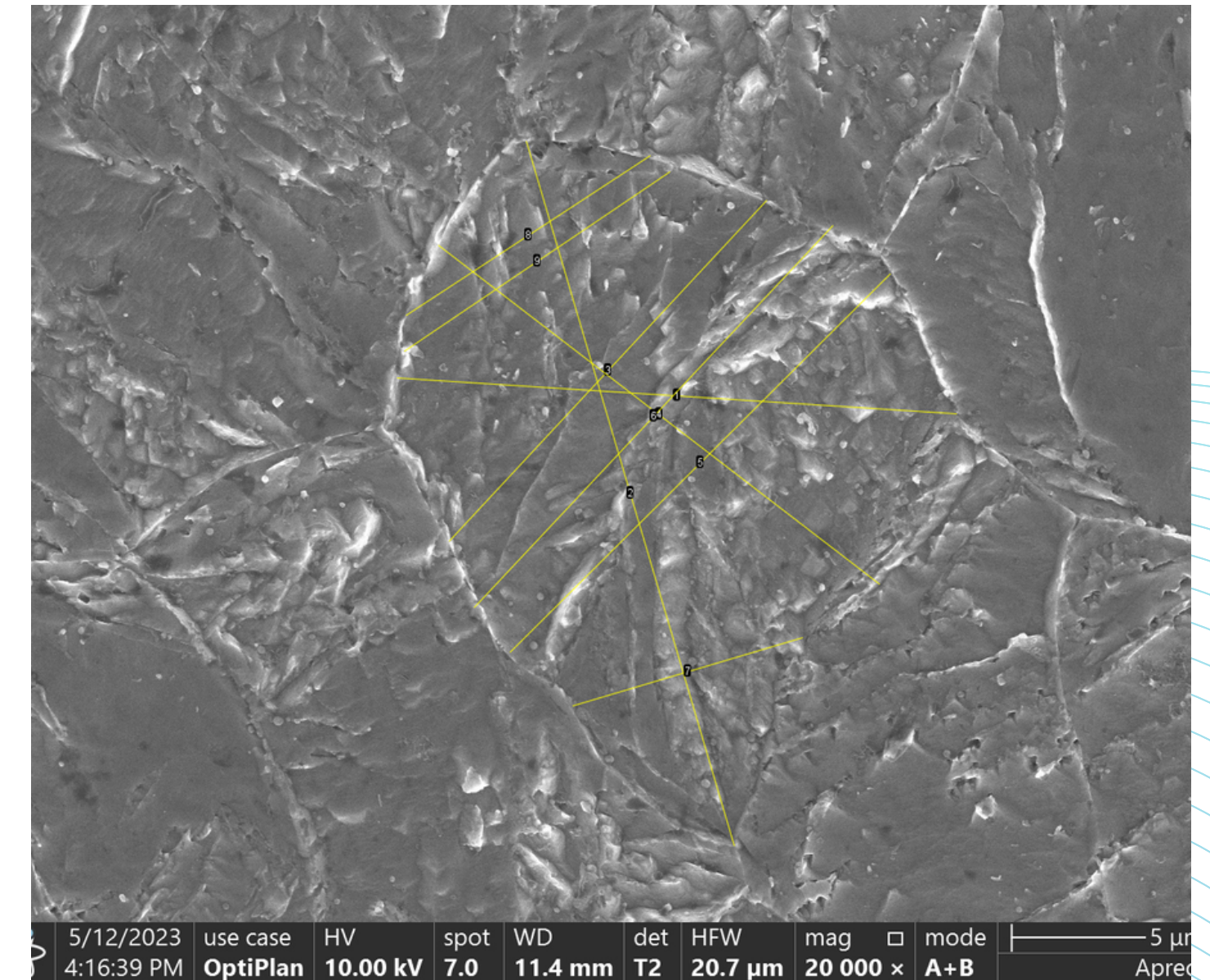
In [1]: import numpy as np
import pandas as pd

In [2]: df=pd.read_csv(r"C:\Users\omiiit\Downloads\image_measurements_100x_4.csv")

In [3]: df.mean(axis=0)

Out[3]: Unnamed: 0      480.500000
Area      185.058594
equivalent_diameter    6.056512
orientation    6.516607
MajorAxisLength    9.810819
MinorAxisLength    5.056262
Perimeter    81.046551
MinIntensity    132.660063
MeanIntensity    146.015949
MaxIntensity    161.787278
dtype: float64

```



Avg. diameter obtained from csv file

- The average diameter calculated from the csv file that the program generated is compared to the average diameter calculated from the grain size of the FESEM image.
- The avg. diameter obtained from the csv file is **6.05 micrometer** shown in the fig 4.a
- The avg. diameter of FESEM image is **6.137 micrometer** calculates using the ImageJ software as shown in fig 4.b

Avg. diameter calculated using ImageJ

Variation of Temperature and Avg. Grain size

1. IN MATERIALS SCIENCE AND METALLURGY, THE FLUCTUATION OF GRAIN SIZE WITH TEMPERATURE IS A KEY TOPIC. THE CONCEPT OF RECRYSTALLIZATION AND GRAIN GROWTH CAN BE USED TO GENERALIZE THE RELATIONSHIP BETWEEN GRAIN SIZE AND TEMPERATURE

RECRYSTALLIZATION

- TRIGGERED BY RISING TEMPERATURES.
- INCREASED TEMPERATURE LEADS TO ENERGY GROWTH WITHIN THE DEFORMED STRUCTURE.
- RESULTS IN THE FORMATION OF NEW, STRAIN-FREE GRAINS.
- TEMPERATURE INFLUENCE
- HIGHER TEMPERATURES GENERALLY ENCOURAGE RECRYSTALLIZATION. THIS PHENOMENON TENDS TO PRODUCE SMALLER GRAIN SIZES.

GRAIN GROWTH PROCESS:

- OCCURS WHEN A MATERIAL IS KEPT AT HIGH TEMPERATURES FOR AN EXTENDED PERIOD.
- ATOMS DIFFUSE OVER GRAIN BOUNDARIES, CAUSING CONSOLIDATION OF NEARBY GRAINS.
- RESULTS IN THE EXPANSION OF ALREADY-EXISTING GRAINS, LEADING TO LARGER GRAIN SIZES.

HIGHER TEMPERATURES

- GENERALLY ENCOURAGE RECRYSTALLIZATION.
- CAUSE STRAIN-FREE GRAINS TO NUCLEATE AND DEVELOP IN SMALLER GRAINS.
- EXTENDED HIGH-TEMPERATURE EXPOSURE:
- ATOMS DIFFUSE ACROSS GRAIN BOUNDARIES DURING PROLONGED EXPOSURE.
- ADJACENT GRAINS COMBINE, CAUSING GRAIN EXPANSION.
- RESULTS IN AN INCREASE IN GRAIN DIAMETERS.

Conclusion

IN THIS STUDY, THE AVERAGE GRAIN SIZE WAS EVALUATED FROM MICROSTRUCTURE IMAGES USING PYTHON AND MACHINE LEARNING .. OPENCV GAVE THE CSV FILE WHICH WAS THEN USED TO OBTAIN THE AVG GRAIN SIZE AND THE SAME FILE COULD BE USED IN A MACHINE LEARNING MODEL TO OBTAIN RESULTS FOR SAME TYPE OF MICROSTRUCTURE IMAGES. AFTER ALL THE TASKS , THE FOLLOWING CONCLUSIONS WERE OBTAINED:

- (1) THE PYTHON CODE COULD BE A BETTER OPTION FOR FINDING AVG GRAIN SIZE RATHER THAN ANY OTHER EXPERIMENTAL PROCEDURES WHICH NEARLY GAVE SAME RESULTS.**
- (2) THE CSV FILE OBTAINED ALSO GAVE OTHER IMPORTANT PARAMETERS OF STUDY OTHER THAN THE GRAIN SIZE.**
- (3) MACHINE LEARNING MODELS COULD BE TRAINED ON THE CSV FILE DATA OBTAINED AND COULD BE USED TO PREDICT AVG GRAIN SIZE OF SIMILAR MICROSTRUCTURE (OPTICAL) IMAGES.**
- (4) PYTHON PROGRAM COULD SAVE US THE USE OF FESEM FOR THE AVERAGE GRAIN SIZE DETERMINATION UP TO SOME EXTENT.**

THANK YOU!

